SYNTHETIC DEMO DOCUMENT · not a real institutional document, no patient data

UZ Gent paediatric haemato-oncology · Local SOP: Fever and neutrope

Version 2.1 · synthetic document generated for a hackathon demo · no real patient or institutional content

1 Scope and purpose

This standard operating procedure (SOP) applies to children and adolescents treated in the paediatric haemato-oncology unit who present with fever during chemotherapy-induced neutropenia. It defines the local work-up, the empirical antimicrobial choices and the escalation rules. Patients enrolled in a clinical trial follow the trial protocol wherever the protocol gives an instruction; this SOP fills the gaps.

Fever is defined as a single oral temperature of 38.5 °C or above, or 38.0 °C sustained for one hour. Neutropenia is defined as an absolute neutrophil count below 0.5 × 10?/L, or below 1.0 × 10?/L and expected to fall.

2 Initial assessment (within 60 minutes)

2.1 Clinical assessment: vital signs, perfusion, mucositis, central line site, perianal region, skin and lungs. 2.2 Investigations: full blood count with differential, CRP, blood cultures from every lumen of the central venous catheter and one peripheral set, urine culture, chest X-ray only when respiratory signs are present.

2.3 Risk stratification: high risk when any of the following applies ? expected neutropenia longer than 7 days, haemodynamic instability, mucositis grade 3 or more, induction phase of acute leukaemia, relapsed disease, or age under 12 months.

3 Empirical antimicrobial therapy

3.1 First line (all risk groups): piperacillin-tazobactam 100 mg/kg (piperacillin component) every 8 hours, maximum 4 g per dose, started within 60 minutes of presentation. 3.2 Add amikacin 15 mg/kg once daily when the patient is haemodynamically unstable or has a known ESBL colonisation. 3.3 Add vancomycin only for catheter-related infection with cellulitis, known MRSA colonisation, or severe mucositis with viridans streptococci risk.

3.4 Persistent fever at 96 hours in a high-risk patient: start empirical antifungal therapy. The local first choice is voriconazole 9 mg/kg twice daily (loading) then 8 mg/kg twice daily; liposomal amphotericin B 3 mg/kg once daily is the alternative when azoles are contraindicated. NOTE: review all co-medication for CYP3A4 interactions before starting an azole.

3.5 Stop rule: in a clinically stable low-risk patient with negative cultures at 48 hours, antibiotics may be stopped before neutrophil recovery, with daily review.

4 Transfusion thresholds and supportive care

4.1 Platelet transfusion threshold during febrile neutropenia: 20 × 10?/L; 50 × 10?/L before lumbar puncture or any invasive procedure. 4.2 Red cell transfusion: haemoglobin below 7 g/dL, or below 8 g/dL with symptoms. 4.3 G-CSF is not given routinely; it may be considered in a high-risk patient with documented infection after discussion with the attending consultant.

4.4 Document every deviation from an active clinical trial protocol in the patient file and inform the trial coordinator within 24 hours. Version 2.1, approved by the unit's infection committee (synthetic).